

SET 1

```
create table person (driver_id int primary key, name varchar(7), address varchar(7));
```

Table created.

```
SQL> create table car (license varchar(10) primary key, model varchar(7), year number);
```

Table created.

```
SQL> create table accident (report_no int primary key, date_acc date, location varchar(8));
```

Table created.

```
SQL> create table owns (driver_id int, license varchar(10), primary key(driver_id, license), foreign key  
(driver_id) references person(driver_id), foreign key (license) references car(license));
```

Table created.

```
SQL> create table participated (driver_id int, model varchar(7), report_no int, damage_amount  
number);
```

Table created.

```
1] SQL> insert into person values (&driver_id, '&name', '&address');
```

Enter value for driver_id: 1

Enter value for name: Amit

Enter value for address: Mumbai

old 1: insert into person values (&driver_id, '&name', '&address')

new 1: insert into person values (1, 'Amit', 'Mumbai')

1 row created.

SQL> /

Enter value for driver_id: 2

Enter value for name: Senha

Enter value for address: Pune

old 1: insert into person values (&driver_id, '&name', '&address')

new 1: insert into person values (2, 'Senha', 'Pune')

1 row created.

SQL> /

Enter value for driver_id: 3

Enter value for name: Rahul

Enter value for address: Nashik

old 1: insert into person values (&driver_id, '&name', '&address')

new 1: insert into person values (3, 'Rahul ', 'Nashik')

1 row created.

SQL> /

Enter value for driver_id: 4

Enter value for name: Mansi

Enter value for address: Nagpur

old 1: insert into person values (&driver_id, '&name', '&address')

new 1: insert into person values (4, 'Mansi', 'Nagpur')

1 row created.

SQL> /

Enter value for driver_id: 5

Enter value for name: Prachi

Enter value for address: Thane

old 1: insert into person values (&driver_id, '&name', '&address')

new 1: insert into person values (5, 'Prachi', 'Thane')

1 row created.

SQL> insert into car values('&license', '&model', &year);

Enter value for license: MH02AB1234

Enter value for model: swift

Enter value for year: 2020

old 1: insert into car values('&license', '&model', &year)

new 1: insert into car values('MH02AB1234', 'swift', 2020)

1 row created.

SQL> /

Enter value for license: MH12CD5678

Enter value for model: baleno

Enter value for year: 2021

old 1: insert into car values('&license', '&model', &year)

new 1: insert into car values('MH12CD5678', 'baleno', 2021)

1 row created.

SQL> /

Enter value for license: MH14EH9012

Enter value for model: i10

Enter value for year: 2022

old 1: insert into car values('&license', '&model', &year)

new 1: insert into car values('MH14EH9012', 'i10', 2022)

1 row created.

SQL> /

Enter value for license: MH15GH3456

Enter value for model: i20

Enter value for year: 2023

old 1: insert into car values('&license', '&model', &year)

new 1: insert into car values('MH15GH3456', 'i20', 2023)

1 row created.

SQL> /

Enter value for license: MH19IJ7890

Enter value for model: creta

Enter value for year: 2024

old 1: insert into car values('&license', '&model', &year)

new 1: insert into car values('MH19IJ7890', 'creta', 2024)

1 row created.

SQL> insert into accident values (&report_no, to_date('&date_acc'), '&location');

Enter value for report_no: 101

Enter value for date_acc: 10-jan-2025

Enter value for location: mumbai

old 1: insert into accident values (&report_no, to_date('&date_acc'), '&location')

new 1: insert into accident values (101, to_date('10-jan-2025'), 'mumbai')

1 row created.

SQL> /

Enter value for report_no: 102

Enter value for date_acc: 10-feb-2025

Enter value for location: pune

old 1: insert into accident values (&report_no, to_date('&date_acc'), '&location')

new 1: insert into accident values (102, to_date('10-feb-2025'), 'pune')

1 row created.

SQL> /

Enter value for report_no: 103

Enter value for date_acc: 15-mar-2025

Enter value for location: nashik

old 1: insert into accident values (&report_no, to_date('&date_acc'), '&location')

new 1: insert into accident values (103, to_date('15-mar-2025'), 'nashik')

1 row created.

SQL> /

Enter value for report_no: 104

Enter value for date_acc: 20-apr-2025

Enter value for location: nagpur

old 1: insert into accident values (&report_no, to_date('&date_acc'), '&location')

new 1: insert into accident values (104, to_date('20-apr-2025'), 'nagpur')

1 row created.

SQL> /

Enter value for report_no: 105

Enter value for date_acc: 25-may-2025

Enter value for location: thane

old 1: insert into accident values (&report_no, to_date('&date_acc'), '&location')

new 1: insert into accident values (105, to_date('25-may-2025'), 'thane')

1 row created.

```
SQL> insert into participated values(&driver_id, '&model', &report_no, &damage_amount);
```

Enter value for driver_id: 1

Enter value for model: swift

Enter value for report_no: 101

Enter value for damage_amount: 15000

```
old 1: insert into participated values(&driver_id, '&model', &report_no, &damage_amount)
```

```
new 1: insert into participated values(1, 'swift', 101, 15000)
```

1 row created.

```
SQL> /
```

Enter value for driver_id: 2

Enter value for model: baleno

Enter value for report_no: 102

Enter value for damage_amount: 10000

```
old 1: insert into participated values(&driver_id, '&model', &report_no, &damage_amount)
```

```
new 1: insert into participated values(2, 'baleno', 102, 10000)
```

1 row created.

```
SQL> /
```

Enter value for driver_id: 3

Enter value for model: i10

Enter value for report_no: 103

Enter value for damage_amount: 20000

```
old 1: insert into participated values(&driver_id, '&model', &report_no, &damage_amount)
```

```
new 1: insert into participated values(3, 'i10', 103, 20000)
```

1 row created.

SQL> /

Enter value for driver_id: 4

Enter value for model: i20

Enter value for report_no: 104

Enter value for damage_amount: 19000

old 1: insert into participated values(&driver_id, '&model', &report_no, &damage_amount)

new 1: insert into participated values(4, 'i20', 104, 19000)

1 row created.

SQL> /

Enter value for driver_id: 5

Enter value for model: creta

Enter value for report_no: 105

Enter value for damage_amount: 25000

old 1: insert into participated values(&driver_id, '&model', &report_no, &damage_amount)

new 1: insert into participated values(5, 'creta', 105, 25000)

1 row created.

2] SQL> desc person;

Name	Null?	Type

DRIVER_ID	NOT NULL	NUMBER(38)
NAME		VARCHAR2(7)
ADDRESS		VARCHAR2(7)

SQL> desc car;

Name	Null?	Type

LICENSE	NOT NULL VARCHAR2(10)
MODEL	VARCHAR2(7)
YEAR	NUMBER

SQL> desc accident;

Name	Null?	Type

REPORT_NO	NOT NULL	NUMBER(38)
DATE_ACC		DATE
LOCATION		VARCHAR2(8)

SQL> desc owns;

Name	Null?	Type

DRIVER_ID	NOT NULL	NUMBER(38)
LICENSE	NOT NULL	VARCHAR2(10)

SQL> desc participated;

Name	Null?	Type

DRIVER_ID		NUMBER(38)
MODEL		VARCHAR2(7)
REPORT_NO		NUMBER(38)
DAMAGE_AMOUNT		NUMBER

3] SQL> select *from person;

DRIVER_ID	NAME	ADDRESS

1	Amit	Mumbai
---	------	--------

2	Senha	Pune
---	-------	------

3 Rahul Nashik

4 Mansi Nagpur

5 Prachi Thane

4] SQL> alter table participated add constraint fk_report foreign key (report_no) references accident(report_no);

Table altered.

5] SQL> alter table participated add constraint chk_damage check(damage_amount >= 0);

Table altered.

6] SQL> alter table person modify name varchar(7) not null;

Table altered.

SQL> desc person;

Name	Null?	Type

DRIVER_ID	NOT NULL	NUMBER(38)
NAME	NOT NULL	VARCHAR2(7)
ADDRESS		VARCHAR2(7)

7] SQL> rename car to vehicle;

Table renamed.

SQL> select *from vehicle;

LICENSE	MODEL	YEAR
---------	-------	------

```
-----  
MH02AB1234 swift      2020  
MH12CD5678 baleno     2021  
MH14EH9012 i10        2022  
MH15GH3456 i20        2023  
MH19IJ7890 creta      2024
```

8] SQL> create table new_person as select *from person;

Table created.

SQL> select *from new_person;

```
DIVER_ID NAME  ADDRESS  
-----  
1 Amit  Mumbai  
2 Senha  Pune  
3 Rahul  Nashik  
4 Mansi  Nagpur  
5 Prachi Thane
```

SQL> drop table new_person;

Table dropped.

SET 2

```
SQL>create table customer (c_id int primary key, cname varchar(8), city varchar(12));
```

Table created.

```
SQL>create table account (c_id int, acc_type varchar(8), amount number, foreign key (c_id)
references customer(c_id));
```

Table created.

```
SQL> INSERT INTO Customer VALUES (1, 'John', 'Nashik');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (2, 'Seema', 'Aurangabad');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (3, 'Amita', 'Nagar');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (4, 'Rakesh', 'Pune');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (5, 'Samata', 'Nashik');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (6, 'Ankita', 'Chandwad');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (7, 'Bhavika', 'Pune');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (8, 'Deepa', 'Mumbai');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (9, 'Nitin', 'Nagpur');
```

1 row created.

```
SQL> INSERT INTO Customer VALUES (10, 'Pooja', 'Pune');
```

1 row created.

```
SQL> INSERT INTO Account VALUES (1, 'Current', 5000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (2, 'Saving', 20000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (3, 'Saving', 70000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (4, 'Saving', 50000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (6, 'Current', 35000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (7, 'Loan', 30000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (8, 'Saving', 50000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (9, 'Saving', 90000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (10, 'Loan', 8000);
```

1 row created.

```
SQL> INSERT INTO Account VALUES (10, 'Current', 45000);
```

1 row created.

```
SQL> select *from customer;
```

C_ID	CNAME	CITY
------	-------	------

```

-----
1 John   Nashik
2 Seema  Aurangabad
3 Amita  Nagar
4 Rakesh Pune
5 Samata Nashik
6 Ankita Chandwad
7 Bhavika Pune
8 Deepa  Mumbai
9 Nitin  Nagpur
10 Pooja Pune

```

10 rows selected.

SQL> select *from account;

```

C_ID ACC_TYPE  AMOUNT
-----

```

```

1 Current    5000
2 Saving     20000
3 Saving     70000
4 Saving     50000
6 Current    35000
7 Loan       30000
8 Saving     50000
9 Saving     90000
10 Loan      8000
10 Current   45000

```

10 rows selected.

1] SQL> select c.cname, a.acc_type, a.amount from customer c inner join account a on c.c_id= a.c_id where a.acc_type='Saving';

CNAME	ACC_TYPE	AMOUNT
-------	----------	--------

Seema	Saving	20000
-------	--------	-------

Amita	Saving	70000
-------	--------	-------

Rakesh	Saving	50000
--------	--------	-------

Deepa	Saving	50000
-------	--------	-------

Nitin	Saving	90000
-------	--------	-------

OR

SQL> select cname, acc_type, amount from customer natural join account where acc_type='Saving';

CNAME	ACC_TYPE	AMOUNT
-------	----------	--------

Seema	Saving	20000
-------	--------	-------

Amita	Saving	70000
-------	--------	-------

Rakesh	Saving	50000
--------	--------	-------

Deepa	Saving	50000
-------	--------	-------

Nitin	Saving	90000
-------	--------	-------

2] SQL> select *from customer natural join account;

C_ID	CNAME	CITY	ACC_TYPE	AMOUNT
------	-------	------	----------	--------

1	John	Nashik	Current	5000
---	------	--------	---------	------

2	Seema	Aurangabad	Saving	20000
---	-------	------------	--------	-------

3	Amita	Nagar	Saving	70000
---	-------	-------	--------	-------

4	Rakesh	Pune	Saving	50000
---	--------	------	--------	-------

6	Ankita	Chandwad	Current	35000
---	--------	----------	---------	-------

7	Bhavika	Pune	Loan	30000
---	---------	------	------	-------

8	Deepa	Mumbai	Saving	50000
---	-------	--------	--------	-------

9	Nitin	Nagpur	Saving	90000
10	Pooja	Pune	Loan	8000
10	Pooja	Pune	Current	45000

10 rows selected.

SQL> select *from customer c left join account a on c.c_id = a.c_id;

C_ID	CNAME	CITY	C_ID	ACC_TYPE	AMOUNT
1	John	Nashik	1	Current	5000
2	Seema	Aurangabad	2	Saving	20000
3	Amita	Nagar	3	Saving	70000
4	Rakesh	Pune	4	Saving	50000
6	Ankita	Chandwad	6	Current	35000
7	Bhavika	Pune	7	Loan	30000
8	Deepa	Mumbai	8	Saving	50000
9	Nitin	Nagpur	9	Saving	90000
10	Pooja	Pune	10	Loan	8000
10	Pooja	Pune	10	Current	45000
5	Samata	Nashik			

11 rows selected.

SQL> select *from customer c right join account a on c.c_id = a.c_id;

C_ID	CNAME	CITY	C_ID	ACC_TYPE	AMOUNT
1	John	Nashik	1	Current	5000
2	Seema	Aurangabad	2	Saving	20000
3	Amita	Nagar	3	Saving	70000

4	Rakesh	Pune	4	Saving	50000
6	Ankita	Chandwad	6	Current	35000
7	Bhavika	Pune	7	Loan	30000
8	Deepa	Mumbai	8	Saving	50000
9	Nitin	Nagpur	9	Saving	90000
10	Pooja	Pune	10	Loan	8000
10	Pooja	Pune	10	Current	45000

10 rows selected.

3] SQL> select *from customer where city = (select city from customer where cname = 'Pooja');

C_ID	CNAME	CITY
4	Rakesh	Pune
7	Bhavika	Pune
10	Pooja	Pune

4] SQL> select *from account where amount < (select avg(amount) from account);

C_ID	ACC_TYPE	AMOUNT
1	Current	5000
2	Saving	20000
6	Current	35000
7	Loan	30000
10	Loan	8000

5] SQL> select c_id from account where amount = (select max(amount) from account);

C_ID

9

6] SQL> select acc_type, min(amount) from account group by acc_type;

ACC_TYPE MIN(AMOUNT)

Current 5000

Saving 20000

Loan 8000

7] SQL> select amount from account where amount > any (select amount from account where acc_type = 'Saving');

AMOUNT

90000

70000

50000

50000

45000

35000

30000

7 rows selected.

OR

SQL> select amount from account where amount > (select min(amount) from account where acc_type = 'Saving');

AMOUNT

70000

50000

35000

30000

50000

90000

45000

7 rows selected.

SET 3

create table department (dept_id int primary key, dept_name varchar(10), location varchar(10));

Table created.

SQL> create table employee (emp_id int primary key, emp_name varchar(8), city varchar(10), salary number, department number, foreign key (department) references department(dept_id));

Table created.

1] SQL> desc department;

Name	Null?	Type

DEPT_ID	NOT NULL	NUMBER(38)
DEPT_NAME		VARCHAR2(8)
LOCATION		VARCHAR2(10)

SQL> desc employee;

Name	Null?	Type

EMP_ID	NOT NULL	NUMBER(38)
EMP_NAME		VARCHAR2(8)
CITY		VARCHAR2(7)
SALARY		NUMBER
DEPARTMENT		NUMBER

2] SQL> INSERT INTO Department VALUES (101, 'IT', 'Mumbai');

1 row created.

SQL> INSERT INTO Department VALUES (102, 'HR', 'Pune');

1 row created.

```
SQL> INSERT INTO Department VALUES (103, 'Sales', 'Nashik');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (104, 'Finance', 'Nagpur');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (105, 'Marketing', 'Aurangabad');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (106, 'Admin', 'Thane');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (107, 'Support', 'Pune');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (108, 'Testing', 'Mumbai');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (109, 'Security', 'Nagar');
```

1 row created.

```
SQL> INSERT INTO Department VALUES (110, 'Research', 'Delhi');
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (1, 'Neha', 'Mumbai', 30000, 101);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (2, 'Deepak', 'Pune', 35000, 102);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (3, 'Ritesh', 'Nashik', 40000, 103);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (4, 'Meena', 'Nagpur', 28000, 104);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (5, 'Ketan', 'Aurangabad', 45000, 105);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (6, 'Reema', 'Thane', 50000, 106);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (7, 'Sejal', 'Mumbai', 32000, 107);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (8, 'Tejas', 'Pune', 38000, 108);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (9, 'Jeetu', 'Nagar', 41000, 109);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (10, 'Dev', 'Delhi', 47000, 110);
```

1 row created.

```
3] SQL> insert into department values (111, 'design', 'nashik');
```

1 row created.

```
SQL> select *from department;
```

```
DEPT_ID DEPT_NAME LOCATION
```

```
-----
```

```
101 IT      Mumbai
```

```
102 HR      Pune
```

```
103 Sales   Nashik
```

```
104 Finance Nagpur
```

```
105 Marketing Aurangabad
```

```
106 Admin   Thane
```

```
107 Support Pune
```

```
108 Testing Mumbai
```

```
109 Security Nagar
```

```
110 Research Delhi
```

111 design nashik

11 rows selected.

4] SQL> insert into employee (emp_id, emp_name, city, department) values (11, 'ajit', 'nashik', 111);

1 row created.

SQL> select *from employee;

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT
1	Neha	Mumbai	30000	101
2	Deepak	Pune	35000	102
3	Ritesh	Nashik	40000	103
4	Meena	Nagpur	28000	104
5	Ketan	Aurangabad	45000	105
6	Reema	Thane	50000	106
7	Sejal	Mumbai	32000	107
8	Tejas	Pune	38000	108
9	Jeetu	Nagar	41000	109
10	Dev	Delhi	47000	110
11	ajit	nashik	111	

11 rows selected.

5] SQL> update employee set salary = salary + (salary * 0.20);

11 rows updated.

SQL> select *from employee;

EMP_ID EMP_NAME CITY SALARY DEPARTMENT

1	Neha	Mumbai	36000	101
2	Deepak	Pune	42000	102
3	Ritesh	Nashik	48000	103
4	Meena	Nagpur	33600	104
5	Ketan	Aurangabad	54000	105
6	Reema	Thane	60000	106
7	Sejal	Mumbai	38400	107
8	Tejas	Pune	45600	108
9	Jeetu	Nagar	49200	109
10	Dev	Delhi	56400	110
11	ajit	nashik	111	

11 rows selected.

OR

SQL> update employee set salary = salary * 1.20;

11 rows updated.

SQL> select *from employee;

EMP_ID EMP_NAME CITY SALARY DEPARTMENT

1	Neha	Mumbai	43200	101
2	Deepak	Pune	50400	102
3	Ritesh	Nashik	57600	103
4	Meena	Nagpur	40320	104
5	Ketan	Aurangabad	64800	105
6	Reema	Thane	72000	106

7	Sejal	Mumbai	46080	107
8	Tejas	Pune	54720	108
9	Jeetu	Nagar	59040	109
10	Dev	Delhi	67680	110
11	ajit	nashik		111

11 rows selected.

6] SQL> select emp_name, salary * 12 as annual_salary from employee where emp_name like '_e%';

EMP_NAME ANNUAL_SALARY

```

-----
Neha      518400
Deepak    604800
Meena     483840
Ketan     777600
Reema     864000
Sejal     552960
Tejas     656640
Jeetu     708480
Dev       812160

```

9 rows selected.

PRACTICE

SQL> select emp_name, salary * 12 as annual_salary from employee where emp_name like 'e%';

no rows selected

SQL> select emp_name, salary * 12 as annual_salary from employee where emp_name like '__e%';

EMP_NAME ANNUAL_SALARY

Deepak 604800

Meena 483840

Reema 864000

Jeetu 708480

SQL> select emp_name, salary * 12 as annual_salary from employee where emp_name like '%u';

EMP_NAME ANNUAL_SALARY

Jeetu 708480

7] SQL> update employee set city = 'pune' where department = (select dept_id from department where dept_name = 'IT');

1 row updated.

SQL> select *from employee;

EMP_ID EMP_NAME CITY SALARY DEPARTMENT

1 Neha pune 43200 101

2 Deepak Pune 50400 102

3 Ritesh Nashik 57600 103

4 Meena Nagpur 40320 104

5 Ketan Aurangabad 64800 105

6 Reema Thane 72000 106

7 Sejal Mumbai 46080 107

8 Tejas Pune 54720 108

9 Jeetu Nagar 59040 109

10 Dev	Delhi	67680	110
11 ajit	nashik		111

11 rows selected.

PRACTICE

USING IN OPERATOR

```
SQL> UPDATE Employee SET city = 'Pune' WHERE department IN ( SELECT dept_id FROM Department
WHERE dept_name = 'IT');
```

1 row updated.

```
SQL> select *from employee;
```

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT
1	Neha	Pune	43200	101
2	Deepak	Pune	50400	102
3	Ritesh	Nashik	57600	103
4	Meena	Nagpur	40320	104
5	Ketan	Aurangabad	64800	105
6	Reema	Thane	72000	106
7	Sejal	Mumbai	46080	107
8	Tejas	Pune	54720	108
9	Jeetu	Nagar	59040	109
10	Dev	Delhi	67680	110
11	ajit	nashik		111

11 rows selected.

USING DIRECTLY DEPARTMENT ID

```
SQL> UPDATE Employee SET city = 'Pune' WHERE department = 101;
```

1 row updated.

SQL> select *from employee;

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT
1	Neha	Pune	43200	101
2	Deepak	Pune	50400	102
3	Ritesh	Nashik	57600	103
4	Meena	Nagpur	40320	104
5	Ketan	Aurangabad	64800	105
6	Reema	Thane	72000	106
7	Sejal	Mumbai	46080	107
8	Tejas	Pune	54720	108
9	Jeetu	Nagar	59040	109
10	Dev	Delhi	67680	110
11	ajit	nashik		111

11 rows selected.

SET 4

STARTING THREE QUERIES ARE SAME LIKE (1,2,7 QUERIES) FROM SET 3.

4] SQL> delete from employee where salary > 100000;

0 rows deleted.

5] SQL> select emp_name, salary - (salary * 0.05) as salary_after_tax from employee;

EMP_NAME SALARY_AFTER_TAX

Neha	41040
Deepak	47880
Ritesh	54720
Meena	38304
Ketan	61560
Reema	68400
Sejal	43776
Tejas	51984
Jeetu	56088
Dev	64296
ajit	

11 rows selected.

6] SQL> select *from employee where city != 'Mumbai' and emp_name like 'a%';

EMP_ID EMP_NAME CITY SALARY DEPARTMENT

11	ajit	nashik	111	
----	------	--------	-----	--

PRACTICE

```
SQL> select *from employee where city != 'Mumbai' and emp_name like 'A%';
```

no rows selected

```
SQL> select *from employee where city <> 'Mumbai' and emp_name like 'a%';
```

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT
11	ajit	nashik	111	

```
QL> select *from employee where not city = 'Mumbai' and emp_name like 'a%';
```

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT
11	ajit	nashik	111	

```
7] SQL> select *from employee e inner join department d on e.department = d.dept_id where  
d.dept_name = 'IT' and e.salary > 50000;
```

no rows selected

PRACTICE

```
SQL> select *from employee e inner join department d on e.department = d.dept_id where  
d.dept_name = 'IT' and e.salary > 40000;
```

EMP_ID	EMP_NAME	CITY	SALARY	DEPARTMENT	DEPT_ID	DEPT_NAME
1	Neha	Pune	43200	101	101	IT

Mumbai

SET 6

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE Departments (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(30) NOT NULL,  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

Table created.

```
SQL> INSERT INTO Departments VALUES (10, 'IT', 101, 1001);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (20, 'HR', 102, 1002);
```


1 row created.

```
SQL> INSERT INTO Departments VALUES (30, 'Sales', 103, 1003);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
2 VALUES (101, 'Vaishali', 'Patil', 'vaishali@gmail.com',  
3 '9876543210', '10-JAN-2025', 'IT_PROG', 50000, 0.10, 201, 10);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
2 VALUES (102, 'Amit', 'Sharma', 'amit@gmail.com',  
3 '9876543211', '10-FEB-2025', 'HR_MGR', 40000, 0.05, 202, 20);
```

1 row created.

```
SQL>
```

```
SQL> INSERT INTO Employees
```

```
2 VALUES (103, 'Sneha', 'Patil', 'sneha@gmail.com',  
3 '9876543212', '15-FEB-2025', 'SA_REP', 45000, 0.08, 203, 10);
```

1 row created.

```
1] SQL> create table EmpTest as select *from Employees;
```

Table created.

2] SQL> INSERT INTO EmpTest (employee_id, first_name, last_name, email, hire_date, job_id, salary)
VALUES (104, USER, 'Patil', 'user@gmail.com', SYSDATE, 'IT_PROG', 30000);

1 row created.

3] SQL> UPDATE EmpTest SET salary = salary + (salary * 0.15) WHERE first_name = 'Vaishali';

1 row updated.

4] SQL> UPDATE EmpTest SET salary = salary + (salary * 0.10) WHERE department_id IN (SELECT
department_id FROM Departments WHERE location_id = 1001);

2 rows updated.

5] SQL> DELETE FROM EmpTest WHERE commission_pct IS NOT NULL;

3 rows deleted.

SQL> select *from EmpTest;

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
104	SYS	Patil	user@gmail.com		12-MAY-26	IT_PROG	30000			

6] SQL> DELETE FROM EmpTest WHERE department_id = 10;

0 rows deleted.

7] SQL> ROLLBACK;

Rollback complete.

SQL> SELECT * FROM EmpTest WHERE department_id = 10;

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
101	Vaishali	Patil	vaishali@gmail.com	9876543210	10-JAN-25	IT_PROG	50000	.1	201	10
103	Sneha	Patil	sneha@gmail.com	9876543212	15-FEB-25	SA_REP	45000	.08	203	10

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
101	Vaishali	Patil	vaishali@gmail.com	9876543210	10-JAN-25	IT_PROG	50000	.1	201	10
103	Sneha	Patil	sneha@gmail.com	9876543212	15-FEB-25	SA_REP	45000	.08	203	10

SET 7

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> INSERT INTO Employees  
2 VALUES (101, 'Vaishali', 'Patil', 'vaishali@gmail.com',  
3 '9876543210', SYSDATE, 'IT_PROG',  
4 50000, 0.10, 201, 10);
```

1 row created.

```
SQL> INSERT INTO Employees  
2 VALUES (102, 'Amit', 'Sharma', 'amit@gmail.com',  
3 '9876543211', SYSDATE, 'HR_MGR',  
4 40000, 0.05, 202, 20);
```

1 row created.

1] SQL> INSERT INTO Employees

2 VALUES (103, 'Sneha', 'Patil', 'sneha@gmail.com',
3 '9876543212', SYSDATE, 'SA_REP',
4 45000, 0.08, 203, 30);

1 row created.

SQL> COMMIT;

Commit complete.

2] SQL> UPDATE Employees SET salary = salary + 5000 WHERE employee_id = 101;

1 row updated.

SQL> ROLLBACK;

Rollback complete.

3] SQL> SAVEPOINT SP1;

Savepoint created.

SQL> UPDATE Employees SET salary = salary + 2000 WHERE employee_id = 102;

1 row updated.

SQL> ROLLBACK TO SP1;

Rollback complete.

4] SQL> SAVEPOINT SP1;

Savepoint created.

SQL> UPDATE Employees SET salary = salary + 1000 WHERE employee_id = 101;

1 row updated.

SQL> SAVEPOINT SP2;

Savepoint created.

SQL> UPDATE Employees SET salary = salary + 2000 WHERE employee_id = 102;

1 row updated.

SQL> ROLLBACK TO SP2;

Rollback complete.

5] SQL> DELETE FROM Employees WHERE employee_id = 102;

1 row deleted.

SQL> COMMIT;

Commit complete.

SET 8

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> INSERT INTO Employees  
VALUES (101, 'Vaishali', 'Patil', 'vaishali@gmail.com',  
'9876543210', SYSDATE, 'IT_PROG', 50000, 0.10, 201, 10);
```

1 row created.

```
SQL> INSERT INTO Employees  
VALUES (102, 'Amit', 'Sharma', 'amit@gmail.com',  
'9876543211', SYSDATE, 'HR_MGR', 40000, 0.05, 202, 20);
```

1 row created.

```
SQL> INSERT INTO Employees  
VALUES (103, 'Sneha', 'Patil', 'sneha@gmail.com',
```

```
'9876543212', SYSDATE, 'SA_REP', 45000, 0.08, 203, 10);
```

1 row created.

```
1] SQL> SELECT MAX(salary) AS highest_salary,  
              MIN(salary) AS lowest_salary,  
              SUM(salary) AS total_salary,  
              AVG(salary) AS average_salary  
              FROM Employees;
```

```
HIGHEST_SALARY LOWEST_SALARY TOTAL_SALARY AVERAGE_SALARY
```

```
-----  
50000      40000      135000      45000
```

```
2] SQL> SELECT job_id,  
              COUNT(*) AS total_people  
              FROM Employees  
              GROUP BY job_id;
```

```
JOB_ID  TOTAL_PEOPLE
```

```
-----  
IT_PROG          1  
HR_MGR           1  
SA_REP           1
```

```
3] SQL> SELECT COUNT(DISTINCT manager_id) AS "Number of Managers" FROM Employees;
```

```
Number of Managers
```

```
-----
```



```
4] SQL> SELECT MAX(salary) - MIN(salary)
        AS DIFFERENCE
        FROM Employees;
```

DIFFERENCE

10000

```
5] SQL> SELECT manager_id,
        MIN(salary) AS lowest_salary
        FROM Employees
        GROUP BY manager_id;
```

MANAGER_ID LOWEST_SALARY

201	50000
202	40000
203	45000

SET 12

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE Departments (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(30) NOT NULL,  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

Table created.

```
SQL> INSERT INTO Departments VALUES (10, 'Sales', 101, 1001);  
1 row created.
```

```
SQL> INSERT INTO Departments VALUES (20, 'HR', 102, 1002);  
1 row created.
```

```
SQL> INSERT INTO Departments VALUES (30, 'IT', 103, 1003);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (101, 'King', 'Sharma', 'king@gmail.com',  
'9876543210', SYSDATE, 'MANAGER', 80000, 0.10, NULL, 10);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (102, 'Amit', 'Patil', 'amit@gmail.com',  
'9876543211', SYSDATE, 'SALESMAN', 40000, 0.05, 101, 10);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (103, 'Sneha', 'Patil', 'sneha@gmail.com',  
'9876543212', SYSDATE, 'CLERK', 30000, 0.02, 101, 20);
```

1 row created.

```
1] SQL> SELECT employee_id, first_name, salary, department_name, location_id  
FROM Employees NATURAL JOIN Departments;
```

EMPLOYEE_ID	FIRST_NAME	SALARY	DEPARTMENT_NAME
-------------	------------	--------	-----------------

LOCATION_ID

102	Amit	40000	Sales
-----	------	-------	-------

1001

```
2] SQL> SELECT E.job_id, E.manager_id, E.salary, E.commission_pct, D.department_name  
FROM Employees E
```

```

JOIN Departments D
ON E.department_id = D.department_id
WHERE E.job_id = 'SALESMAN';

```

JOB_ID	MANAGER_ID	SALARY	COMMISSION_PCT	DEPARTMENT_NAME

SALESMAN	101	40000	.05	Sales

```

3] SQL> SELECT E.first_name, E.job_id, E.department_id, D.department_name
      FROM Employees E
      JOIN Departments D
      ON E.department_id = D.department_id
      WHERE D.location_id = 1001;

```

FIRST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME

King	MANAGER	10	Sales
Amit	SALESMAN	10	Sales

```

4] SQL> SELECT employee_id, first_name, manager_id
      FROM Employees
      ORDER BY employee_id;

```

EMPLOYEE_ID	FIRST_NAME	MANAGER_ID

101	King	
102	Amit	101
103	Sneha	101

SET 13

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE Departments (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(30) NOT NULL,  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE SALGRADE (  
    grade NUMBER,  
    losal NUMBER,  
    hisal NUMBER  
);
```

Table created.

```
SQL> INSERT INTO SALGRADE VALUES (1, 1000, 2000);
```

1 row created.

```
SQL> INSERT INTO SALGRADE VALUES (2, 2001, 4000);
```

1 row created.

```
SQL> INSERT INTO SALGRADE VALUES (3, 4001, 6000);
```

1 row created.

```
SQL> INSERT INTO SALGRADE VALUES (4, 6001, 8000);
```

1 row created.

```
SQL> INSERT INTO SALGRADE VALUES (5, 8001, 10000);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (10, 'Sales', 101, 1001);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (20, 'HR', 102, 1002);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (30, 'IT', 103, 1003);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (101, 'Amit', 'Patil', 'amit@gmail.com',
```

```
'9876543210', SYSDATE, 'MANAGER',
```

```
5000, 0.10, 201, 10);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (102, 'Sneha', 'Sharma', 'sneha@gmail.com',
'9876543211', SYSDATE, 'CLERK',
3000, 0.05, 202, 20);
```

1 row created.

SQL> INSERT INTO Employees

```
VALUES (103, 'Rohit', 'Patel', 'rohit@gmail.com',
'9876543212', SYSDATE, 'ANALYST',
7000, 0.08, 203, 30);
```

1 row created.

1] SQL> DESC SALGRADE;

Name	Null?	Type

GRADE		NUMBER
LOSAL		NUMBER
HISAL		NUMBER

```
SQL> SELECT E.first_name, E.job_id, D.department_name, E.salary, S.grade
FROM Employees E
JOIN Departments D
ON E.department_id = D.department_id
JOIN SALGRADE S
ON E.salary BETWEEN S.losal AND S.hisal;
```

FIRST_NAME	JOB_ID	DEPARTMENT_NAME	SALARY

GRADE			

Amit	MANAGER	Sales	5000

Sneha	CLERK	HR	3000
2			

Rohit	ANALYST	IT	7000
4			

2] SQL> SELECT E.first_name, E.salary, E.manager_id, D.department_name
 FROM Employees E
 JOIN Departments D
 ON E.department_id = D.department_id
 WHERE E.salary > 2000;

FIRST_NAME	SALARY	MANAGER_ID	DEPARTMENT_NAME

Amit	5000	201	Sales
Sneha	3000	202	HR
Rohit	7000	203	IT

3] SQL> SELECT E.employee_id, E.first_name, E.job_id, E.department_id, D.department_name,
 D.location_id
 FROM Employees E
 LEFT OUTER JOIN Departments D
 ON E.department_id = D.department_id;

EMPLOYEE_ID	FIRST_NAME	JOB_ID	DEPARTMENT_ID

DEPARTMENT_NAME	LOCATION_ID		

101 Amit	MANAGER	10	
Sales	1001		

102 Sneha	CLERK	20
HR	1002	
103 Rohit	ANALYST	30
IT	1003	

4] SQL> SELECT E.first_name,
D.department_name
FROM Employees E
RIGHT OUTER JOIN Departments D
ON E.department_id = D.department_id;

FIRST_NAME	DEPARTMENT_NAME

Amit	Sales
Sneha	HR
Rohit	IT

5] SQL> SELECT E.employee_id, D.department_name, D.location_id
FROM Employees E
FULL OUTER JOIN Departments D
ON E.department_id = D.department_id;

EMPLOYEE_ID	DEPARTMENT_NAME	LOCATION_ID

101 Sales	1001	
102 HR	1002	
103 IT	1003	

SET 14

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE Departments (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(30) NOT NULL,  
    manager_id NUMBER(6),  
    location_id VARCHAR2(20)  
);
```

Table created.

```
SQL> INSERT INTO Departments VALUES (10, 'Sales', 101, 'Pune');  
1 row created.
```

```
SQL> INSERT INTO Departments VALUES (20, 'HR', 102, 'Mumbai');  
1 row created.
```

```
SQL> INSERT INTO Departments VALUES (30, 'IT', 103, 'Delhi');
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (40, 'Marketing', 104, 'Pune');
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (101, 'Amit', 'Patil', 'amit@gmail.com',  
'9876543210', SYSDATE, 'SALESMAN',  
5000, 0.10, 201, 10);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (102, 'Sneha', 'Sharma', 'sneha@gmail.com',  
'9876543211', SYSDATE, 'CLERK',  
3000, 0.05, 202, 20);
```

1 row created.

```
SQL> INSERT INTO Employees
```

```
VALUES (103, 'Rohit', 'Patel', 'rohit@gmail.com',  
'9876543212', SYSDATE, 'MANAGER',  
7000, 0.08, 203, 30);
```

1 row created.

```
1] SQL> SELECT E.job_id,
```

```
E.manager_id,
```

```
E.salary,
```

```

        E.commission_pct,
        D.department_name
FROM Employees E
JOIN Departments D
ON E.department_id = D.department_id
WHERE E.job_id = 'SALESMAN';

```

JOB_ID	MANAGER_ID	SALARY	COMMISSION_PCT	DEPARTMENT_NAME
SALESMAN	201	5000	.1	Sales

2] SQL> SELECT E.first_name,
 E.job_id,
 E.department_id,
 D.department_name
 FROM Employees E
 JOIN Departments D
 ON E.department_id = D.department_id
 WHERE D.location_id = 'Pune';

FIRST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME
Amit	SALESMAN	10	Sales

3] SQL> SELECT E.first_name,
 D.department_name
 FROM Employees E
 RIGHT OUTER JOIN Departments D
 ON E.department_id = D.department_id;

FIRST_NAME	DEPARTMENT_NAME
------------	-----------------

```
-----
```

Amit	Sales
Sneha	HR
Rohit	IT
	Marketing

```
4] SQL> SELECT E.first_name,
           D.department_name
FROM Departments D
LEFT OUTER JOIN Employees E
ON E.department_id = D.department_id;
```

```
FIRST_NAME      DEPARTMENT_NAME
-----
```

Amit	Sales
Sneha	HR
Rohit	IT
	Marketing

```
5] SQL> SELECT E.employee_id,
           E.first_name,
           D.location_id
FROM Employees E
JOIN Departments D
ON E.department_id = D.department_id
WHERE E.job_id = 'CLERK';
```

```
EMPLOYEE_ID FIRST_NAME      LOCATION_ID
-----
```

102	Sneha	Mumbai
-----	-------	--------

SET 23

```
SQL> CREATE TABLE Employees (  
    employee_id NUMBER(6) PRIMARY KEY,  
    first_name VARCHAR2(20),  
    last_name VARCHAR2(25) NOT NULL,  
    email VARCHAR2(25) NOT NULL,  
    phone_number VARCHAR2(20),  
    hire_date DATE NOT NULL,  
    job_id VARCHAR2(10) NOT NULL,  
    salary NUMBER(8,2),  
    commission_pct NUMBER(2,2),  
    manager_id NUMBER(6),  
    department_id NUMBER(4)  
);
```

Table created.

```
SQL> CREATE TABLE Departments (  
    department_id NUMBER(4) PRIMARY KEY,  
    department_name VARCHAR2(30) NOT NULL,  
    manager_id NUMBER(6),  
    location_id NUMBER(4)  
);
```

Table created.

```
SQL> INSERT INTO Employees  
VALUES (101, 'ALLEN', 'Patil', 'allen@gmail.com',  
'9876543210', SYSDATE, 'CLERK',  
3000, 0.05, 201, 10);
```

1 row created.

SQL> INSERT INTO Employees

```
VALUES (102, 'MARTIN', 'Sharma', 'martin@gmail.com',  
'9876543211', SYSDATE, 'MANAGER',  
6000, 0.10, 202, 20);
```

1 row created.

SQL> INSERT INTO Employees

```
VALUES (103, 'KING', 'Patel', 'king@gmail.com',  
'9876543212', SYSDATE, 'PRESIDENT',  
9000, 0.15, NULL, 30);
```

1 row created.

SQL> INSERT INTO Employees

```
VALUES (104, 'BLAKE', 'Joshi', 'blake@gmail.com',  
'9876543213', SYSDATE, 'SALESMAN',  
5000, 0.08, 203, 20);
```

1 row created.

SQL> INSERT INTO Employees

```
VALUES (105, 'JONES', 'Kale', 'jones@gmail.com',  
'9876543214', SYSDATE, 'ANALYST',  
4000, 0.07, 204, 10);
```

1 row created.

SQL> INSERT INTO Departments VALUES (10, 'Sales', 101, 1001);

1 row created.

```
SQL> INSERT INTO Departments VALUES (20, 'HR', 102, 1002);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (30, 'IT', 103, 1003);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (40, 'Admin', 104, 1004);
```

1 row created.

```
SQL> INSERT INTO Departments VALUES (50, 'Marketing', 105, 1005);
```

1 row created.

```
1] SQL> SELECT first_name, salary
        FROM Employees
        WHERE salary > (
            SELECT salary
            FROM Employees
            WHERE first_name = 'ALLEN'
        );
```

FIRST_NAME	SALARY
MARTIN	6000
KING	9000

BLAKE	5000
JONES	4000

```
2] SQL> SELECT first_name, job_id
FROM Employees
WHERE job_id = (
    SELECT job_id
    FROM Employees
    WHERE first_name = 'MARTIN'
);
```

FIRST_NAME	JOB_ID

MARTIN	MANAGER

```
3] SQL> SELECT first_name, department_id
FROM Employees
WHERE department_id <> (
    SELECT department_id
    FROM Employees
    WHERE first_name = 'KING'
);
```

FIRST_NAME	DEPARTMENT_ID

ALLEN	10
MARTIN	20
BLAKE	20
JONES	10

```
4] SQL> SELECT first_name, salary
```

```

FROM Employees
WHERE salary > (
    SELECT salary
    FROM Employees
    WHERE first_name = 'BLAKE'
);

```

FIRST_NAME	SALARY
MARTIN	6000
KING	9000

```

5] SQL> SELECT first_name, hire_date
FROM Employees
WHERE hire_date > (
    SELECT hire_date
    FROM Employees
    WHERE first_name = 'JONES'
);

```

no rows selected

```

6] SQL> SELECT first_name, department_id
FROM Employees
WHERE department_id = (
    SELECT department_id
    FROM Employees
    WHERE first_name = 'CLARK'
);

```

no rows selected

SET 24

SAME TABLE LIKE SET 23

```
1] SQL> SELECT first_name, salary
FROM Employees
WHERE salary > ANY (
    SELECT salary
    FROM Employees
    WHERE department_id = 30
);
```

no rows selected

```
2] SQL> SELECT first_name, salary
FROM Employees
WHERE salary > ALL (
    SELECT salary
    FROM Employees
    WHERE department_id = 20
);
```

FIRST_NAME	SALARY

KING	9000

```
3] SQL> SELECT first_name, department_id
FROM Employees
WHERE department_id IN (
    SELECT department_id
    FROM Employees
    WHERE job_id = 'MANAGER'
);
```

FIRST_NAME	DEPARTMENT_ID
------------	---------------

MARTIN	20
--------	----

BLAKE	20
-------	----

4] SQL> SELECT E.first_name,

E.salary,

E.department_id

FROM Employees E

WHERE E.salary > (

SELECT AVG(salary)

FROM Employees

WHERE department_id = E.department_id

);

FIRST_NAME	SALARY	DEPARTMENT_ID
------------	--------	---------------

MARTIN	6000	20
--------	------	----

JONES	4000	10
-------	------	----

5] SQL> SELECT E.first_name,

E.salary,

E.department_id

FROM Employees E

WHERE E.salary = (

SELECT MAX(salary)

FROM Employees

WHERE department_id = E.department_id

);

FIRST_NAME	SALARY	DEPARTMENT_ID
------------	--------	---------------

MARTIN	6000	20
KING	9000	30
JONES	4000	10

```
6] SQL> SELECT first_name, job_id
FROM Employees
WHERE job_id IN (
    SELECT job_id
    FROM Employees
    WHERE department_id = 20
);
```

FIRST_NAME	JOB_ID
------------	--------

MARTIN	MANAGER
BLAKE	SALESMAN

SET 9

```
SQL> CREATE TABLE Employee (  
    emp_id NUMBER PRIMARY KEY,  
    emp_name VARCHAR(10) NOT NULL,  
    department VARCHAR(5) NOT NULL,  
    city VARCHAR(6),  
    salary NUMBER CHECK (salary > 0)  
);
```

Table created.

```
SQL> INSERT INTO Employee VALUES (101, 'Amit', 'IT', 'Pune', 70000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (102, 'Sneha', 'HR', 'Mumbai', 50000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (103, 'Rohit', 'IT', 'Pune', 65000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (104, 'Neha', 'Sales', 'Delhi', 45000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (105, 'Vaishali', 'HR', 'Mumbai', 55000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (106, 'Karan', 'IT', 'Pune', 80000);
```

1 row created.

```
SQL> INSERT INTO Employee VALUES (107, 'Pooja', 'Sales', 'Delhi', 62000);
```

1 row created.

SQL> select *from employee;

EMP_ID	EMP_NAME	DEPAR	CITY	SALARY
101	Amit	IT	Pune	70000
102	Sneha	HR	Mumbai	50000
103	Rohit	IT	Pune	65000
104	Neha	Sales	Delhi	45000
105	Vaishali	HR	Mumbai	55000
106	Karan	IT	Pune	80000
107	Pooja	Sales	Delhi	62000

7 rows selected.

1] SQL> SELECT department,
SUM(salary) AS total_salary
FROM Employee
GROUP BY department;

DEPAR	TOTAL_SALARY
IT	215000
HR	105000
Sales	107000

2] SQL> SELECT department,
AVG(salary) AS average_salary
FROM Employee
GROUP BY department;

DEPAR AVERAGE_SALARY

IT 71666.6667

HR 52500

Sales 53500

```
3] SQL> SELECT city,
      COUNT(*) AS employee_count
      FROM Employee
      GROUP BY city
      HAVING COUNT(*) > 3;
```

no rows selected

```
4] SQL> SELECT department,
      MAX(salary) AS maximum_salary
      FROM Employee
      GROUP BY department
      HAVING MAX(salary) > 60000;
```

DEPAR MAXIMUM_SALARY

IT 80000

Sales 62000

```
5] SQL> SELECT department,
      city,
      SUM(salary) AS total_salary
      FROM Employee
      GROUP BY department, city;
```


DEPAR CITY TOTAL_SALARY

IT Pune 215000

HR Mumbai 105000

Sales Delhi 107000

6] SQL> SELECT department,

COUNT(*) AS total_employees

FROM Employee

GROUP BY department

HAVING COUNT(*) > 1;

DEPAR TOTAL_EMPLOYEES

IT 3

HR 2

Sales 2

7] SQL> SELECT city,

COUNT(*) AS employee_count

FROM Employee

GROUP BY city

ORDER BY employee_count DESC;

CITY EMPLOYEE_COUNT

Pune 3

Delhi 2

Mumbai 2

SET 10

SAME TABLE AS SET 9

```
1] SQL> SELECT department,
      SUM(salary) AS total_salary
      FROM Employee
      GROUP BY department
      HAVING SUM(salary) > 200000;
```

DEPAR TOTAL_SALARY

IT	215000
----	--------

```
2] SQL> SELECT department,
      AVG(salary) AS average_salary
      FROM Employee
      GROUP BY department
      HAVING AVG(salary) > 40000;
```

DEPAR AVERAGE_SALARY

IT	71666.6667
----	------------

HR	52500
----	-------

Sales	53500
-------	-------

```
3] SQL> SELECT city,
      COUNT(*) AS employee_count
      FROM Employee
      GROUP BY city;
```

CITY EMPLOYEE_COUNT

Pune	3
Mumbai	2
Delhi	2

4] SQL> SELECT department,
MAX(salary) AS maximum_salary
FROM Employee
GROUP BY department;

DEPAR MAXIMUM_SALARY

IT	80000
HR	55000
Sales	62000

5] SQL> SELECT city,
AVG(salary) AS average_salary
FROM Employee
GROUP BY city;

CITY AVERAGE_SALARY

Pune	71666.6667
Mumbai	52500
Delhi	53500

6] SQL> SELECT department,
COUNT(*) AS employee_count
FROM Employee
GROUP BY department
HAVING COUNT(*) > 3;

no rows selected

```
7] SQL> SELECT city,  
COUNT(*) AS employee_count  
FROM Employee  
GROUP BY city  
ORDER BY employee_count ASC;
```

CITY EMPLOYEE_COUNT

Mumbai	2
--------	---

Delhi	2
-------	---

Pune	3
------	---

SET 11

1] SQL> CREATE TABLE dept1 (deptno INTEGER PRIMARY KEY, dname VARCHAR2(30) NOT NULL,
loc VARCHAR2(30) NOT NULL);

Table created.

2] SQL> ALTER TABLE dept1 ADD mgr NUMBER;

Table altered.

3] SQL> CREATE TABLE empl (empno INTEGER, ename VARCHAR2(20), sal NUMBER(10,2), deptno
NUMBER);

Table created.

4] SQL> ALTER TABLE empl
ADD CONSTRAINT emp_pk
PRIMARY KEY (empno);

Table altered.

SQL> ALTER TABLE empl
MODIFY ename VARCHAR2(20) NOT NULL;

Table altered.

SQL> ALTER TABLE empl
ADD CONSTRAINT sal_chk
CHECK (sal > 10000);

Table altered.

SQL> ALTER TABLE empl
ADD CONSTRAINT dept_fk
FOREIGN KEY (deptno)
REFERENCES dept1(deptno);

Table altered.

SQL> DESC dept1;

Name	Null?	Type
DEPTNO	NOT NULL	NUMBER(38)
DNAME	NOT NULL	VARCHAR2(30)
LOC	NOT NULL	VARCHAR2(30)
MGR		NUMBER

SQL> DESC empl;

Name	Null?	Type
EMPNO	NOT NULL	NUMBER(38)
ENAME	NOT NULL	VARCHAR2(20)
SAL		NUMBER(10,2)
DEPTNO		NUMBER

5] SQL> CREATE TABLE dept11 AS SELECT * FROM dept1;

Table created.

SQL> CREATE TABLE emp11 AS SELECT * FROM empl;

Table created.

6] SQL> ALTER TABLE empl ADD address VARCHAR2(30);

Table altered.

7] SQL> ALTER TABLE empl RENAME COLUMN sal TO salary;

Table altered.

8] SQL> SELECT constraint_name,

constraint_type

```
FROM user_constraints
WHERE table_name = 'EMPL';
```

CONSTRAINT_NAME

C

-

DEPT_FK

R

EMP_PK

P

SYS_C008450

C

CONSTRAINT_NAME

C

-

SAL_CHK

C

SET 20

```
SQL> CREATE TABLE student ( S_ID NUMBER PRIMARY KEY, name VARCHAR(8) NOT NULL,  
dept_name VARCHAR(12), tot_cred NUMBER );
```

Table created.

```
SQL> CREATE TABLE instructor ( T_ID NUMBER PRIMARY KEY, name VARCHAR(8) NOT NULL,  
dept_name VARCHAR(12), salary NUMBER );
```

Table created.

```
SQL> CREATE TABLE course ( course_id NUMBER PRIMARY KEY, title VARCHAR(10), dept_name  
VARCHAR(12), credits NUMBER );
```

Table created.

```
SQL> INSERT INTO student VALUES (1, 'Aman', 'Computer', 25);
```

1 row created.

```
SQL> INSERT INTO student VALUES (2, 'Amol', 'IT', 20);
```

1 row created.

```
SQL> INSERT INTO student VALUES (3, 'Samir', 'Computer', 22);
```

1 row created.

```
SQL> INSERT INTO student VALUES (4, 'Ramesh', 'Electronics', 18);
```

1 row created.

```
SQL> INSERT INTO student VALUES (5, 'Amit', 'Computer', 24);
```

1 row created.

```
SQL> INSERT INTO instructor VALUES (101, 'Amit', 'Computer', 50000);
```

1 row created.

```
SQL> INSERT INTO instructor VALUES (102, 'Amol', 'IT', 35000);
```


1 row created.

```
SQL> INSERT INTO instructor VALUES (103, 'Sneha', 'Computer', 60000);
```

1 row created.

```
SQL> INSERT INTO instructor VALUES (104, 'Rohit', 'Electronics', 45000);
```

1 row created.

```
SQL> INSERT INTO instructor VALUES (105, 'Kiran', 'IT', 48000);
```

1 row created.

```
SQL> INSERT INTO course VALUES (201, 'DBMS', 'Computer', 4);
```

1 row created.

```
SQL> INSERT INTO course VALUES (202, 'Java', 'IT', 3);
```

1 row created.

```
SQL> INSERT INTO course VALUES (203, 'Networks', 'Computer', 4);
```

1 row created.

```
SQL> INSERT INTO course VALUES (204, 'Python', 'IT', 3);
```

1 row created.

```
SQL> INSERT INTO course VALUES (205, 'OS', 'Computer', 4);
```

1 row created.

```
SQL> select *from student;
```

S_ID	NAME	DEPT_NAME	TOT_CRED
1	Aman	Computer	25
2	Amol	IT	20

3	Samir	Computer	22
4	Ramesh	Electronics	18
5	Amit	Computer	24

SQL> select *from instructor;

T_ID	NAME	DEPT_NAME	SALARY
101	Amit	Computer	50000
102	Amol	IT	35000
103	Sneha	Computer	60000
104	Rohit	Electronics	45000
105	Kiran	IT	48000

SQL> select *from course;

COURSE_ID	TITLE	DEPT_NAME	CREDITS
201	DBMS	Computer	4
202	Java	IT	3
203	Networks	Computer	4
204	Python	IT	3
205	OS	Computer	4

```
1] SQL> SELECT dept_name
      AVG(salary) AS average_salary
      FROM instructor
      GROUP BY dept_name
      HAVING AVG(salary) > 42000;
```

DEPT_NAME	AVERAGE_SALARY
Computer	55000

Electronics 45000

2] SQL> UPDATE instructor

SET salary = salary + (salary * 0.10)

WHERE dept_name = 'Computer';

2 rows updated.

SQL> select *from instructor;

T_ID	NAME	DEPT_NAME	SALARY
101	Amit	Computer	55000
102	Amol	IT	35000
103	Sneha	Computer	66000
104	Rohit	Electronics	45000
105	Kiran	IT	48000

3] SQL> SELECT name

FROM instructor

WHERE name NOT IN ('Amol', 'Amit');

NAME

Sneha

Rohit

Kiran

4] SQL> SELECT name

FROM student

WHERE LOWER(name) LIKE '%am%';

NAME

Aman

Amol

Samir

Ramesh

Amit

5] SQL> SELECT S.name

FROM student S, course C

WHERE S.dept_name = 'Computer'

AND C.title = 'DBMS'

AND S.dept_name = C.dept_name;

NAME

Aman

Samir

Amit

SET 32

SET 34

SET 35